

Chapter # 01 (Physics and Measurement)

1. Example 81.3

If radius of a sphere is measured as 7.5 cm with error of 0.03 cm. Find the approximate error calculating its volume.

2.

2. Example 81.4

Consider the length of cube is given as 5.75 ± 0.3 cm and you want to find absolute uncertainty in volume.

3. What is the percent uncertainty in the measurement 3.67 ± 0.25 m?

4. A rectangular holding tank 25.0 m in length and 15.0 m in width is used to store water for short period of time in an industrial plant. If 2980 m^3 water is pumped into the tank. What is the depth of the water?

5. A watch factory claims that its watches gain or lose not more than 10 seconds in a year. How accurate is this watch?

9. The diameter of moon is 3480 km. What is the volume of the moon? How many moons would be needed to create a volume equal to that of Earth?

10. Prove that the following equations are dimensionally correct :-

$$(i) f = \frac{1}{2} L \sqrt{\frac{E}{\mu}}, \quad (ii) v = f \lambda$$

$$(iii) T = 2\pi \sqrt{\frac{L}{g}}, \quad (iv) T = 2\pi \sqrt{\frac{m}{k}}$$

$$(v) S = V_i t + \frac{1}{2} a t^2$$

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CHAPTER: 2:-

Numericals:-

1.

1. A helicopter is ascending at the rate of 12 m/s . At a height of 80 m above ground, a package is dropped. How long does the package take to reach the ground?

2. Two tug boats are towing a ship each exerts a force of 6000 N , and the angle between two ropes is 60° . Calculate the resultant force on the ship?

3. A car starts from rest and moves with a constant acceleration. During the 5^{th} second of its motion, it covers a distance of 36 meters . Calculate:

a- acceleration of the car.

b- the total distance covered by the car during this time.

4. At what angle the range of projectile becomes equal to the height of projectile?

5. A mortar shell is fired a ground level target 500 m distance with an initial velocity of 90 m/s . What is the launch angle?

03

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Chapter: 03:-

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Numericals:-

1. A car weighing 9800 N is moving with a speed of 40 km/h . On the application of breaks it comes to rest after travelling a distance of 50 meters . Calculate the average retarding force?
2. A helicopter weighs 3920 N . Calculate the force on it if it is ascending up at the rate of 2 m/s . What will be the force on helicopter if it is moving up with a constant speed of 4 m/s ?
3. A 100 grams bullet is fired from a 10 kg gun with a speed of 100 m/s . What is the speed of recoil of the gun?
4. A machine gun fires 10 bullets per second into a target. Each bullet weighs 20 gram and had a speed of 1500 m/s . Find the necessary force to hold the gun in position?
5. A 50 grams bullet is fired into 10 kg block that is suspended by a long cord so that it can swing as a pendulum. If the block is displaced so that its centre of gravity rises by 10 cm , what was the speed of bullet?
6. A 70-gram ball collides with another ball of mass 140 gram . The initial velocity of the first ball is 9 m/s to right while the second ball is at rest. If the collision were perfectly elastic, what would be the velocity of two balls after the collision?

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7. A truck weighing 2500 kg and moving with velocity of 21 m/s collides with a stationary weighing 1000 kg . The truck and car move together after the impact. Calculate their common velocity?

Worked example: 3-2:-

- * A hockey puck with a mass of 0.2 kg is sliding in the ice at a velocity of 16 m/s . It collides with wall and bounces back with a velocity of -8 m/s . The collision lasts of 0.1 seconds . Calculate the impulse experienced by the hockey puck and the change in its momentum?

Worked example: 3-3:-

- * Two objects, A and B, are initially at rest on a frictionless surface. Object A has a mass of 0.5 kg , and object B has a mass of 0.8 kg . Object A collides with object B. After the collision, object A moves to the right at a velocity of 4 m/s . and object B moves to the left at a velocity of 2 m/s .
- (a) Calculate the total momentum before the collision.
- (b) Calculate the total momentum after the collision.
- (c) Calculate the kinetic energy before and after collision.

Q6

(*) A golf club exerts an average force of 800 N on a golf ball for 0.02 second. If the initial velocity of the ball is 40 m/s and its final velocity is 60 m/s , what is the impulse experienced by the ball? (Self assessment pg: 61 Q: 3)

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Rotational And Circular Motion

Worked example 4.3:

What is the magnitude of force and the centripetal acceleration of a car having mass of 300 kg? Following are a curve of radius 500m at a speed of 100 km/h? also compare the acceleration with that due to gravity for this fairly gentle curve taken at highway speed?

Worked example 4.5:

The International space station orbits at an altitude of 400 km above the surface of earth. What is the space station orbital velocity?

Example Number 4.6:

A basketball spinning on the finger of an athlete has angular velocity $\omega = 120.0$ rad/s. The moment of inertia of a sphere that is hollow, where m is the mass and R is the radius. If the basketball has a weight of 0.600 kg and has

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radius of 0.1200 m , what is the angular momentum of this basketball?

Q: Numerical no 2:

A car accelerates uniformly from the rest and reaches a speed of 22 m/s in 9 s . If the diameter of a tire is 58 cm , find.

- the number of revolution the tire make during this motion, assuming no slipping and
- the final rotational speed of the tire in revolution per second.

Q: Numerical no 3:

An ordinary workshop grindstone has a radius of 7.5 cm and rotates 6500 rev/s

- ~~it~~ calculate the magnitude of centripetal acceleration at its edge in m/s^2 and convert it into multiple of g .
- what is linear speed of a point on the

Q: Numerical no 5:

A satellite wishes to orbit the earth at height of 100 km (approximately 60 miles)

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the surface of earth, Determine the speed, acceleration and orbital period of the satellite (Given: $M_{\text{earth}} = 5.98 \times 10^{24} \text{ kg}$, $R_{\text{earth}} = 6.37 \times 10^6 \text{ m}$).

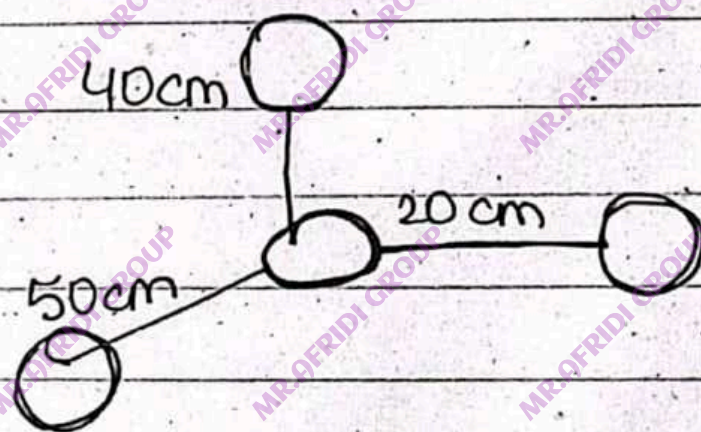
Q: Numerical no 11:

Determine the angular momentum of Earth (a) about its rotation axis (Assume the Earth as uniform sphere), and (b) in its orbit around the sun (Take Earth as a particle orbiting the Sun). The Earth has mass $6 \times 10^{24} \text{ kg}$ and radius ~~6.4~~ $6.4 \times 10^6 \text{ m}$, and is $1.5 \times 10^8 \text{ km}$ from the Sun.

Q What is the ideal banking angle for a gentle turn of 1.20 km radius on a highway with a 105 km/h speed limit (about 65 mi/h), assuming everyone travels at the limit?

Q A 1500 kg car moving on a flat, horizontal road negotiates a curve as shown in figure 4.21. If the radius of the car is 35.0 m and the co-efficient of static friction between the tires and dry pavement is 0.523 , find the maximum speed the car can have and still make the turn successfully.

Q: A system of points shown in figure 4.22. Each particle has same mass of 0.3 kg and they all lie in the same plane. What is the moment of inertia of the system about given axis?



What is the angular momentum of 2.9 kg uniform cylindrical grinding wheel of radius 20 cm. when rotating 1550 rpm? (b) How much torque is required to stop in 6s?

CHAPTER # 05

Important Numericals.

* Example : 5.4

A Person riding their bike has a mass of 120 kg; they are riding at 10 m/s. Suddenly a dog crosses the road and to avoid hitting the dog the bicyclist brakes applying a braking force of 500 N for a distance of 10 meters. What is the final velocity of the bicyclist when they stop braking?

Num : 03

* When an object is thrown upwards it rises to a height 'h'. How high is the object in terms of h when it has lost one third of its original K.E?

Numerical : 04

* A 70 kg man runs up a hill through a height of 3 m in 2 seconds.

(a) How much work does he do against gravitational field? (b) What is the average power output?

Numerical : 05

* A neutron travels a distance of 12 m in a time interval of 3.6×10^{-14} sec. Assuming its speed was constant, find its Kinetic Energy? Take the mass of neutron 1.7×10^{-27} kg.

Numerical : 06

* A stone is thrown vertically upwards and can reach to a height of 10 m, find the speed of stone when it is just 2 m above the ground.

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Numerical : 09

- * A 1 kilowatt motor pump, pumps the water from the ground to a height of 10 m. Find how much liters of water it can pump in one hour?

Numerical : 10

- * A rocket of mass 2 kg is launched in air, when it attains height of 15 m the 400 joule of its chemical fuel burns. Find the speed of rocket at maximum height?

Numerical 11

- * A motor pumps the water at the rate of 500 gram/minute to the height of 120 m. If the motor is 50% efficient then how much input electric power is needed?

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"Chapter no. 06"

Question no. 05

- A flat-bottom river barge is 30 ft wide, 45 ft long and 15 ft deep. (a) How many m^3 of water will it displace while the top stays 1 m above the water? (b) What is its weight in tons in dry dock? The barge contains under these conditions if the empty barge weighs 160 tons in dry dock?

Question no: 06

- A canal lock gate is 20 m wide and 10 m deep. Calculate the thrust acting on it assuming that the water in the canal is in level with the top of the gate. Density of water is 1000 kg/m^3 .

Question no: 07

- A tank 4 m long, 3 m wide and 2 m deep is filled to the brim with paraffin (density 800 kg/m^3). Calculate the pressure on the base? What is the thrust on the base?

Question no: 08

- A rectangular boat is 4.0 m wide, 8.0 m long, and 3.0 m deep. (a) How much water will it displace if the top stays 1.0 m above the water. (b) What load will be the boat contain under these conditions if the empty boat weighs 160 tons in dry dock?

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Question no: 09

- A hot air balloon has a volume of 2200 m^3 . The density of air at temperature of 20°C is 1.205 kg/m^3 . The density of hot air ~~balloon~~ inside the balloon at a temperature of 100°C is 0.946 kg/m^3 . How much weight can the hot air can lift?

Question no: 10

- A spherical balloon has a radius of 7.15 m and is filled with helium. How large a cargo can it lift, assuming that the skin and structure of the balloon have a mass of 930 kg ? Neglect the buoyant force on the cargo volume itself?

Question no: 02.

The pressure in a water pipe in the ground floor of a building is $4 \times 10^5 \text{ Pa}$ but three floors up it is only $2 \times 10^5 \text{ Pa}$. What is the height between the ground floor and the third floor? The water in the pipe may be assumed to be stationary; density of water $= 1 \times 10^3 \text{ kg/m}^3$.

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Example : 6.1

Q: A hydraulic system consists of two connected cylinders. Cylinder A and Cylinder B. Cylinder A has a piston with a radius of 5 cm and Cylinder B has a piston with a radius of 10 cm. A force of 200 N is applied to Cylinder A. Calculate the force exerted by Cylinder B.

Example : 6.2

When a crown of mass 14.7 kg is submerged in water, an accurate scale reads only 13.4 kg. Is the crown made of gold?

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Chapter: 08

Example: 8.3

Three charges $q_1 = +2 \mu\text{C}$, $q_2 = +3 \mu\text{C}$ and $q_3 = +4 \mu\text{C}$ are placed in air at the vertices of an equilateral of sides 10 cm (see figure) calculate the magnitude of the resultant force acting on the charge q_3 ?

Q#1.

- 1) (a) Calculate the value of two equal charges if they repel one another with a force of 0.1 N when situated 50 cm apart in a vacuum.
- (b) What would be the size of the charges if they were situated in an insulating liquid whose permittivity was ten times that of a vacuum?

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Q₂ How far apart must two protons be if the magnitude of the electrostatic force acting on either one due to the other ~~one~~ is equal to the magnitude of the gravitational force on a proton at Earth's surface? (mass of proton = 1.67×10^{-27} kg, charge of proton = 1.6×10^{-19} C (ms 0.119m))

Q₆ Figure shows two charged particles on an x axis: $-q = -3.20 \times 10^{-19}$ C at $x = -3.00$ m and $q = 2.20 \times 10^{-19}$ C at $x = 1.00$ m. What are the (a) magnitude and (b) direction (relative to the positive direction of the x axis) of the net electric field produced at point P at $y = 4.00$ m?

Q₈ The square surface shown in figure measures 3.2 mm on each side. It is immersed in a uniform electric field with magnitude $E = 1800$ N/C and with field lines at an angle of $\theta = 35^\circ$ with a normal to the surface, as shown. Take that normal to be directed "outward" as though the

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surface were one face of a box. Calculate the electric flux through the surface.

Q#9. An electron is liberated from the lower of two large parallel metal plates separated by a distance $h = 20\text{ cm}$. The upper plate has a potential of 2400 volts relative to the lower. How long does the electron take to reach it?

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